

PROJECT REPORT

AI Document Risk & Compliance Classifier

Intelligent Document Risk Analysis, Compliance Detection & Automated
Classification

Submitted by

SHIFANA SHERIN J

Department of Artificial Intelligence and Data Science

K .RAMAKRISHNAN COLLEGE OF ENGINEERING

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1. Project Title

AI Document Risk & Compliance Classifier

Subtitle: An AI-Based System for Automated Document Risk Assessment, Compliance Detection and Intelligent Manual Review

2. Abstract

Organizations handle a large number of business documents such as contracts, policies, reports, invoices, and agreements. Manually reviewing these documents for potential risks and compliance issues is time-consuming and may result in important information being overlooked.

The AI Document Risk & Compliance Classifier is an Artificial Intelligence and Natural Language Processing based system designed to automatically analyze documents and classify them according to their risk and compliance level. The system extracts text from uploaded documents, preprocesses the content, identifies important terms and patterns, and applies machine learning techniques to determine whether a document belongs to a low, medium, or high-risk category.

The system also provides a confidence score for each prediction and sends uncertain documents for manual verification. Important terms and relevant document sections can be highlighted to improve explainability. An interactive dashboard displays document classifications, risk levels, flagged documents, and overall analytics. This system helps reduce manual effort, improve consistency, and support faster and more effective document review.

3. Problem Statement

Organizations process a large number of documents that may contain contractual obligations, sensitive information, compliance requirements, and potential business risks. Manually reviewing every document requires significant time and effort and may lead to inconsistent decisions or missed risk indicators.

Traditional document management systems mainly focus on storing, searching, and retrieving documents rather than intelligently identifying potential risks and compliance concerns. In addition, uncertain documents may not be automatically separated for detailed human review.

Therefore, there is a need for an AI-based document classification system that can automatically analyze document content, identify potential risk and compliance categories, provide confidence levels, highlight important information, and send uncertain cases for manual verification.

4. Objectives

The main objectives of this project are:

- To automatically analyze business documents using AI and NLP techniques.
- To extract and preprocess text from PDF and document files.
- To classify documents according to their risk level.
- To identify potential compliance-related content and risk indicators.
- To use machine learning or NLP features for document classification.
- To provide a confidence score for each classification.
- To identify important terms or sections contributing to the prediction.
- To automatically flag low-confidence or high-risk documents for manual review.
- To detect similar or related documents when required.
- To provide an interactive dashboard for monitoring document risk.
- To maintain document classification history for future analysis.
- To improve the speed and consistency of document review.

5. Proposed System

The proposed system is an AI-powered document analysis and classification system that accepts business documents as input and automatically evaluates their risk and compliance characteristics.

First, the uploaded document is processed using text extraction techniques. Unnecessary formatting and irrelevant content are removed during preprocessing. The cleaned text is then converted into machine-readable features using techniques such as TF-IDF or text embeddings.

A machine learning or NLP-based classifier analyzes these features and assigns the document to an appropriate risk category such as Low Risk, Medium Risk, or High Risk.

The system can also identify compliance-related keywords and important sections.

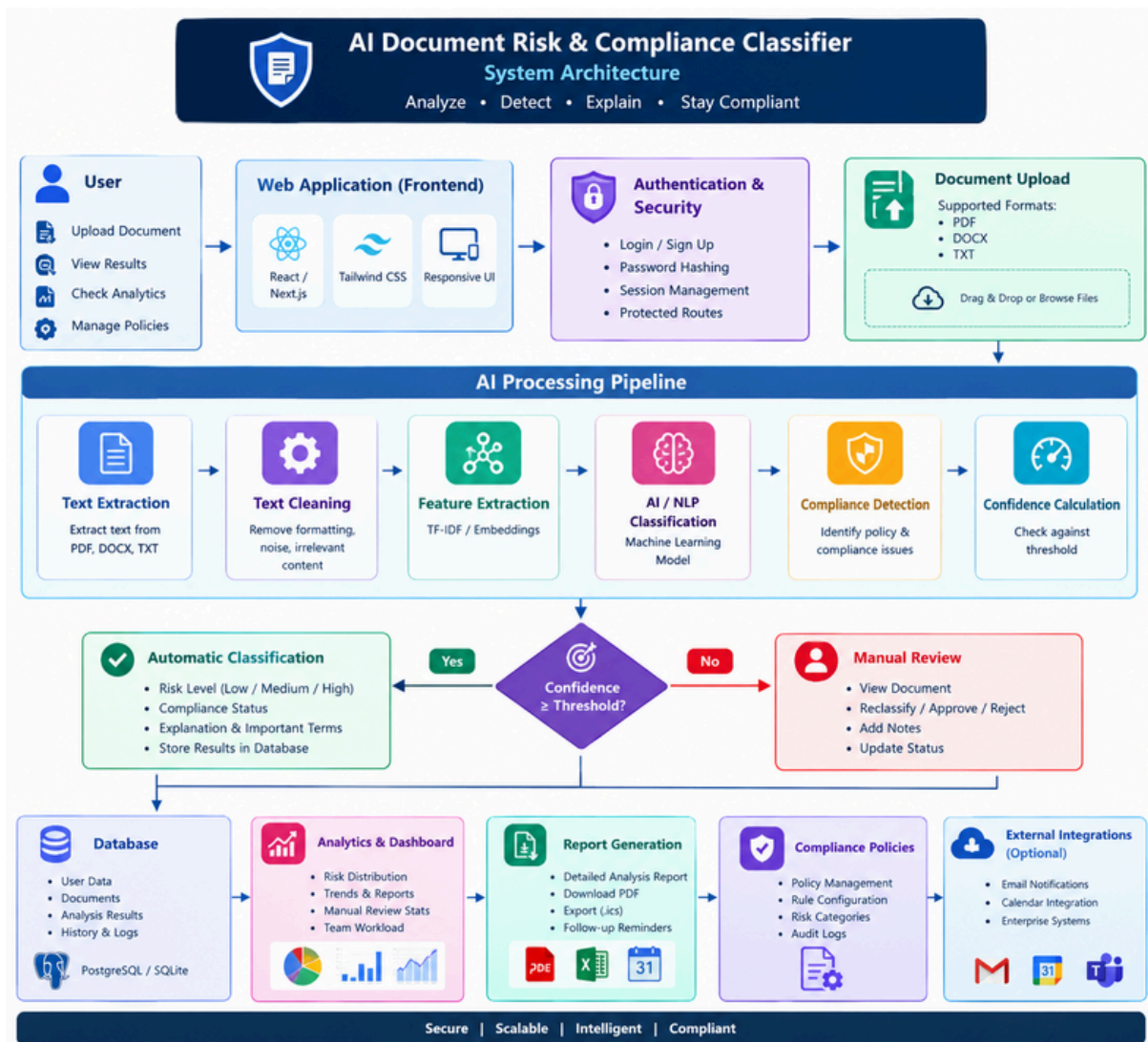
A confidence score is generated along with each prediction. If the confidence is below a predefined threshold, the document is automatically marked for manual review. The system can also provide important terms or similar documents to help users understand the reason behind the classification.

The final results are displayed through an interactive dashboard containing risk classifications, flagged documents, confidence scores, and analytics.

System Workflow

Document Upload → Text Extraction → Text Preprocessing → Feature Extraction → AI/NLP Classification → Risk & Compliance Detection → Confidence Scoring → Manual Review / Classification → Database → Dashboard & Analytics

6. System Architecture



7. Modules Used

1. User Authentication Module

Provides secure login and access control for authorized users.

2. Document Upload Module

Allows users to upload business documents such as PDF and DOCX files.

3. Text Extraction Module

Extracts readable text from uploaded documents for further processing.

4. Text Preprocessing Module

Removes unwanted symbols, formatting, stop words, and unnecessary content.

5. Feature Extraction Module

Converts document text into numerical features using TF-IDF, embeddings, or similar NLP techniques.

6. Risk Classification Module

Uses a machine learning model to classify documents into different risk categories.

7. Compliance Detection Module

Identifies keywords, phrases, or patterns related to compliance requirements and potential risks.

8. Confidence & Validation Module

Generates confidence scores and identifies uncertain or incomplete predictions.

9. Explainability Module

Displays important words, phrases, or document sections that influenced the classification.

10. Manual Review Module

Allows users to inspect, correct, and verify documents flagged by the AI system.

11. Dashboard & Analytics Module

Displays risk distribution, flagged documents, confidence scores, and classification statistics.

12. Database & History Module

Stores document information, predictions, review results, and classification history.

8. Technologies Used

Technology	Purpose
Python	Main programming language
Natural Language Processing (NLP)	Understanding and processing document text
Machine Learning	Risk and compliance classification
Scikit-learn	ML models and feature processing
spaCy / NLTK	Text preprocessing and NLP
TF-IDF	Converting text into numerical features
BERT / Transformers	Advanced contextual text analysis
Pandas	Data processing and analysis
PyPDF / PDF libraries	Extracting text from PDF documents
SQLite / MySQL	Storing document and classification information
Streamlit	Building the interactive dashboard
Plotly / Matplotlib	Creating charts and visual analytics
Git & GitHub	Version control and project manage

9. Key Features

Core Features

- Document Upload – Supports uploading PDF, DOCX, and TXT business documents.
- Automatic Text Extraction – Extracts readable text from uploaded documents for analysis.
- OCR Support – Enables processing of scanned or image-based documents.
- Text Preprocessing – Removes unwanted formatting, symbols, noise, and irrelevant content.
- Natural Language Processing – Understands document content and identifies meaningful words, phrases, and patterns.
- Feature Extraction – Converts document text into useful numerical features using TF-IDF or text embeddings.
- AI-Based Risk Classification – Automatically classifies documents into categories such as Low, Medium, and High Risk.
- Compliance Detection – Identifies terms and content related to organizational policies and compliance requirements.
- Risk Indicator Detection – Detects keywords and patterns that may indicate potential business, contractual, or compliance risks.
- Important Term Identification – Highlights important words and phrases that influence the classification.
- Document Section Analysis – Identifies relevant sections such as obligations, penalties, restrictions, deadlines, and compliance clauses.
- Confidence Score – Provides a confidence percentage for each AI prediction.
- Low-Confidence Detection – Automatically flags uncertain predictions for additional verification.
- High-Risk Document Flagging – Separates high-risk documents so reviewers can prioritize them.
- Manual Review – Allows authorized users to inspect, approve, edit, or reject AI-generated classifications.
- Explainable AI – Provides reasons or important terms behind the predicted risk category.
- Validation Mechanism – Checks for missing, incomplete, or inconsistent document information.
- Duplicate Document Detection – Identifies identical or highly similar documents to avoid unnecessary repeated analysis.
- Document Similarity Analysis – Finds related documents that may have similar content or risk characteristics.

10. Advantages

Reduces Manual Effort – Automatically analyzes and classifies large numbers of documents.

Saves Time – Speeds up the initial document review process.

Early Risk Identification – Helps identify potentially risky documents quickly.

Improves Consistency – Applies the same classification process to documents.

Confidence-Based Review – Sends uncertain predictions for human verification.

Better Compliance Monitoring – Helps identify documents containing compliance-related information.

Explainable Results – Important terms and sections can be shown to support predictions.

Scalable – Can process large collections of documents.

Centralized Dashboard – Provides an easy way to monitor document risk.

Improves Decision Making – Gives reviewers useful information before detailed manual analysis.

LIMITATIONS

Dependent on Document Quality – Poorly scanned or incomplete documents may produce inaccurate results.

Classification Errors – AI models may sometimes classify documents incorrectly.

Limited Context Understanding – Complex legal or business language can be difficult for automated systems.

Training Data Dependency – Model performance depends on the quality and diversity of training data.

False Positives – Some safe documents may be incorrectly flagged as risky.

False Negatives – Some potential risks may not be detected.

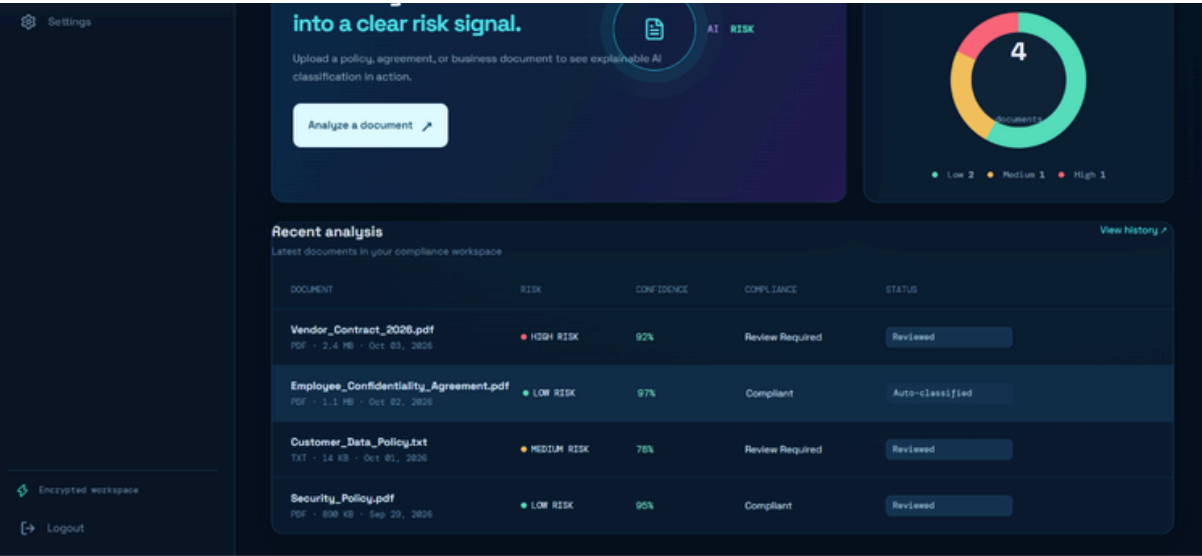
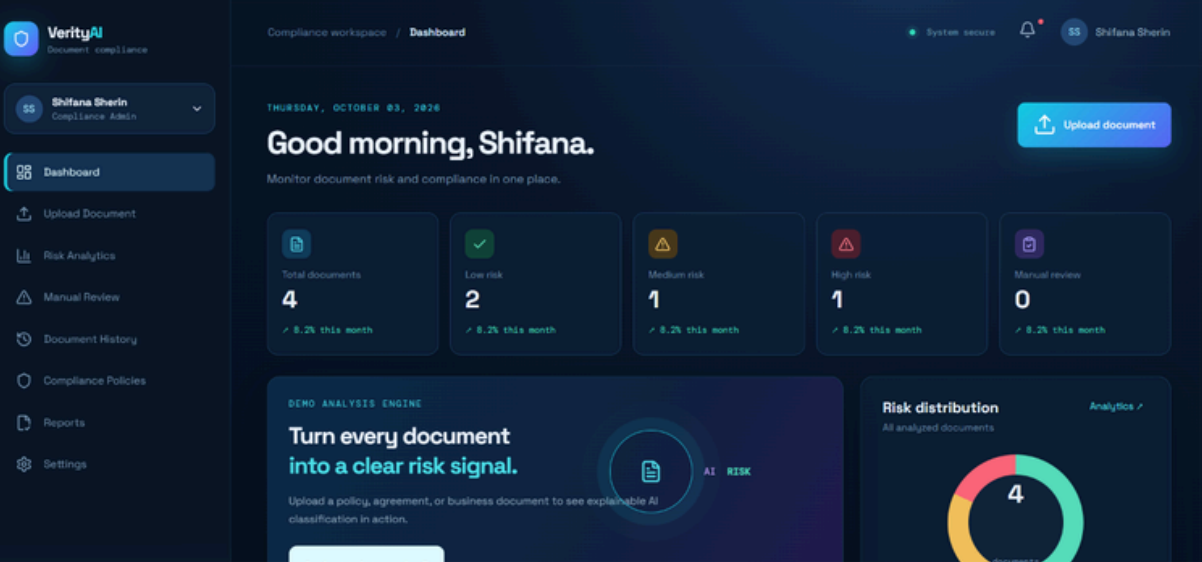
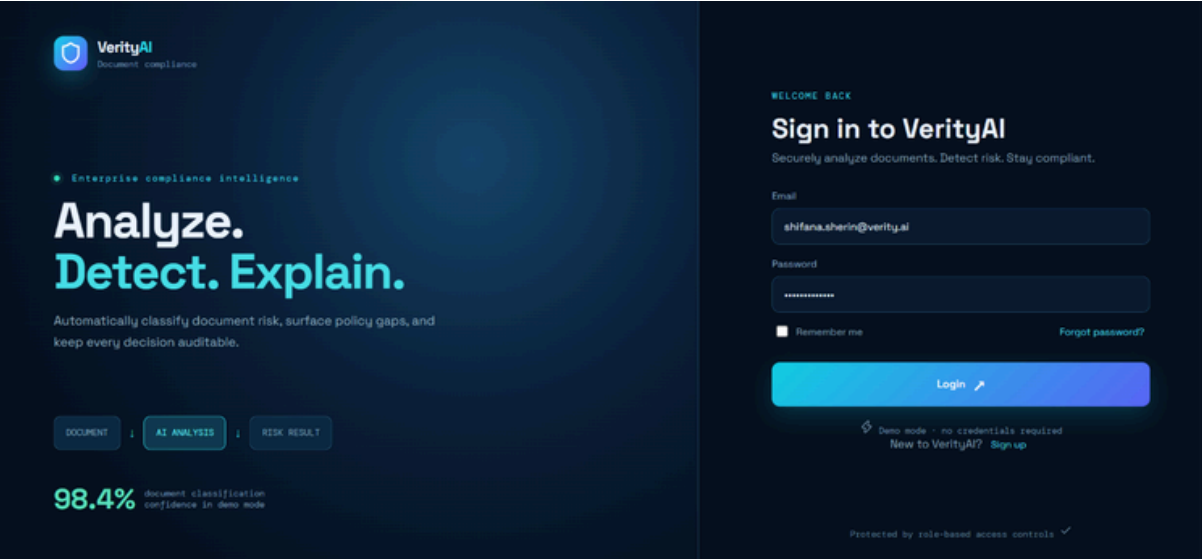
Human Review Required – Critical or low-confidence documents still require expert verification.

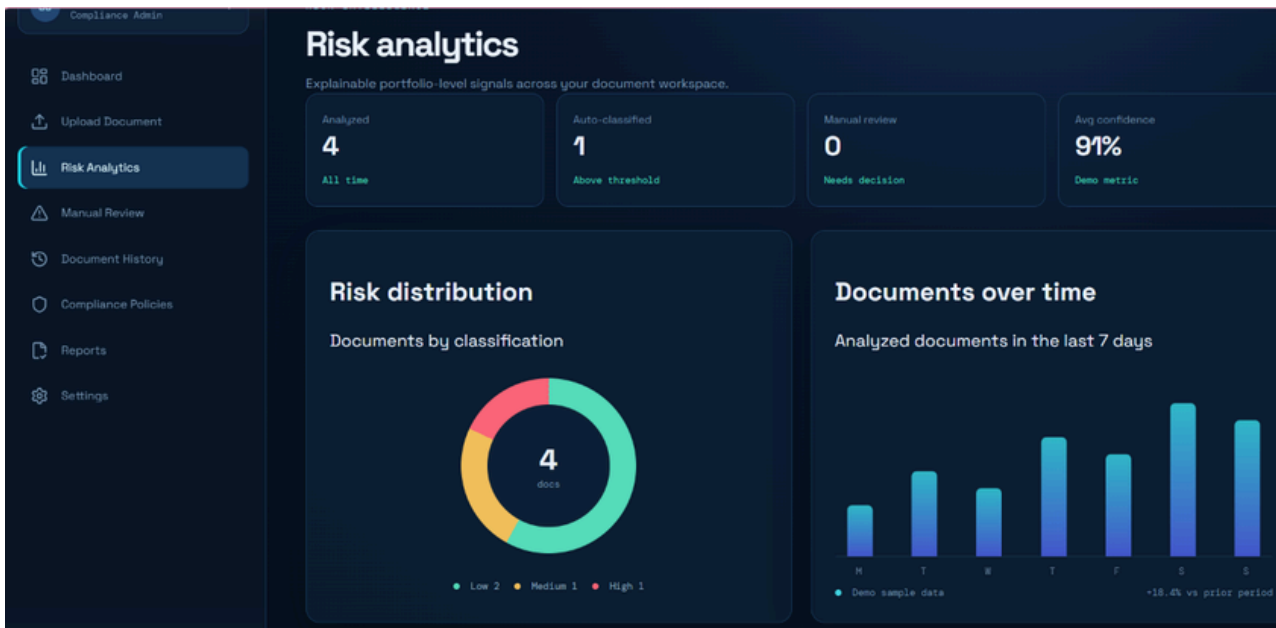
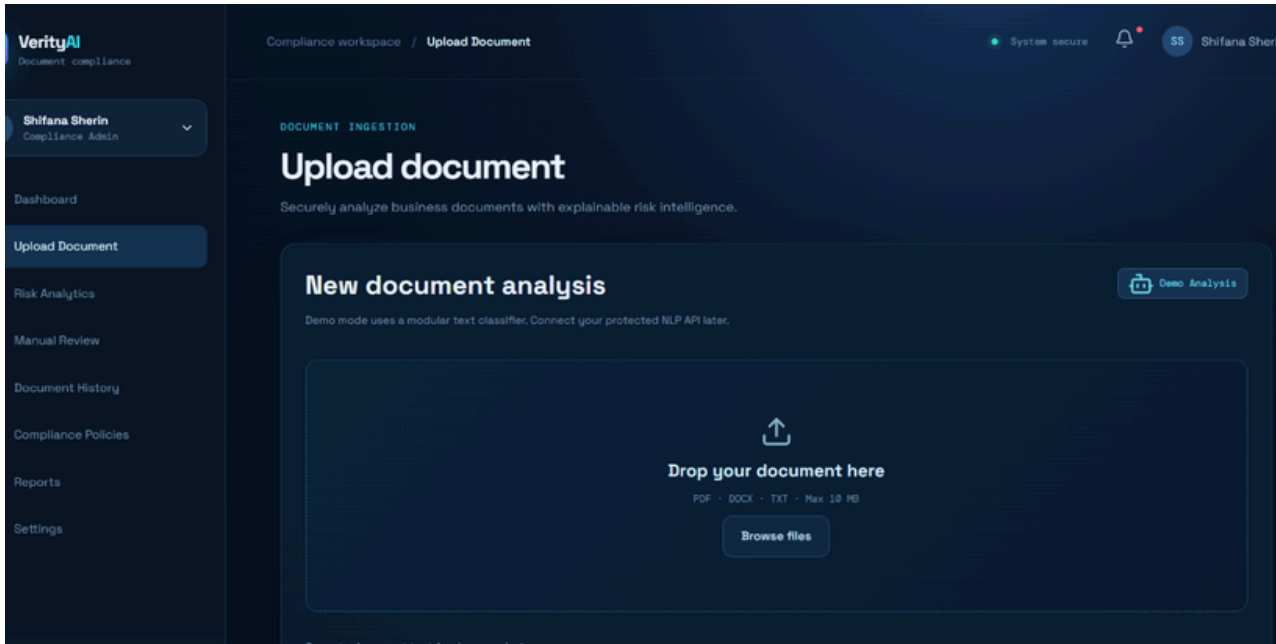
Privacy Concerns – Business documents may contain confidential or sensitive information.

Model Maintenance – The system may need regular updates when policies and compliance requirements change.

Computational Requirements – Advanced NLP models may require higher processing power.

12.Output Screenhot





Document	Risk	Confidence	Compliance Date	Status
Vendor_Co	HIGH	92% Review Req	Oct-03	2026 Reviewed
Employee	LOW	97% Compliant	Oct-02	2026 Auto-classified
Customer_IM	MEDIUM	78% Review Req	Oct-01	2026 Reviewed
Security_Po	LOW	95% Compliant	Sep-29	2026 Reviewed

SSShifana Sherin
Compliance Admin

Dashboard

Upload Document

Risk Analytics

Manual Review

Document History

Compliance Policies

Reports

Settings

Encrypted workspace

Logout

AUDIT TRAIL

Document history

Search, inspect, and manage every analyzed document.

Q Search documents...

Risk filters

DOCUMENT	RISK	CONFIDENCE	COMPLIANCE	DATE	ACTIONS
Vendor_Contract_2026.pdf PDF • 2.4 MB	HIGH RISK	92%	Review Required	Oct 03, 2026	View
Employee_Confidentiality_Agreement.pdf PDF • 1.1 MB	LOW RISK	97%	Compliant	Oct 02, 2026	View
Customer_Data_Policy.txt TXT • 14 KB	MEDIUM RISK	78%	Review Required	Oct 01, 2026	View
Security_Policy.pdf PDF • 880 KB	LOW RISK	95%	Compliant	Sep 29, 2026	View

SSShifana Sherin
Compliance Admin

Dashboard

Upload Document

Risk Analytics

Manual Review

Document History

Compliance Policies

Reports

Settings

CONTROL CENTER

Compliance policies

Define the policy categories used by the demo classifier.

New policy category...

Data Privacy

Risk keywords and review rules for data privacy.

Active

Information Security

Risk keywords and review rules for information security.

Active

Confidentiality

Risk keywords and review rules for confidentiality.

Active

Data Retention

Risk keywords and review rules for data retention.

Active

Third-Party Access

Risk keywords and review rules for third-party access.

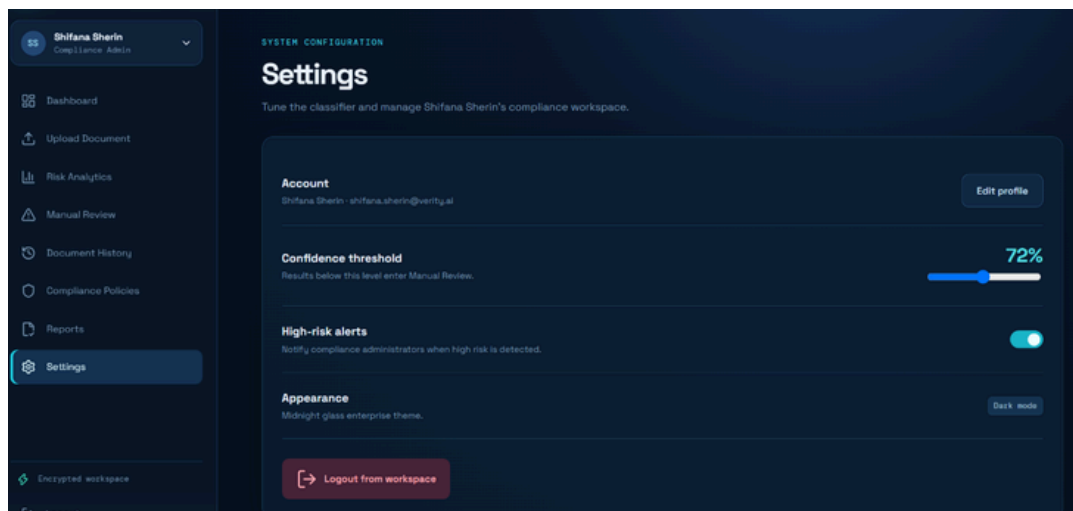
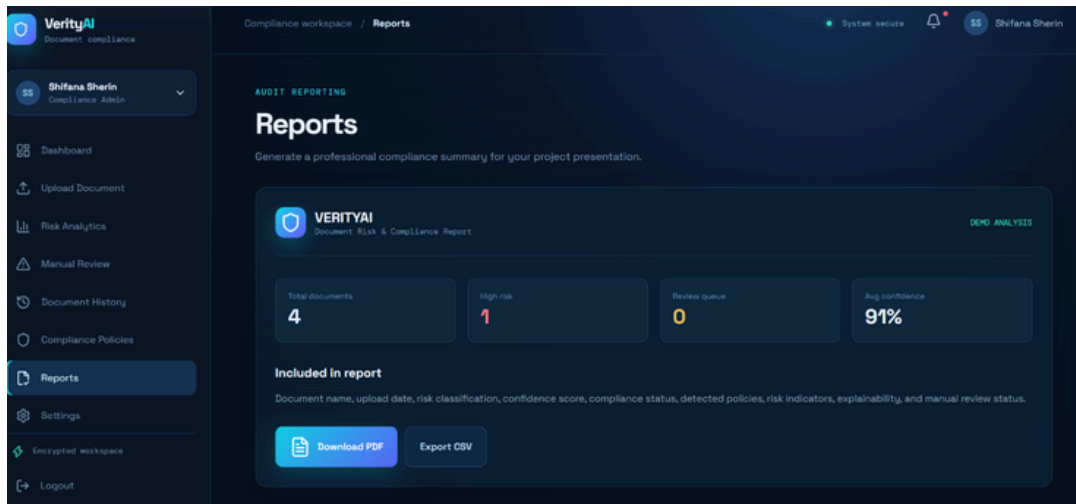
Active

Access Control

Risk keywords and review rules for access control.

Active

+ Add policy



13. Future Scope

- Advanced AI Models – Large Language Models and advanced transformer architectures can improve document understanding.
- Multilingual Support – The system can be extended to analyze documents in multiple languages.
- Real-Time Risk Detection – Documents can be analyzed immediately during upload or editing.
- Automatic Policy Matching – Documents can be compared with organizational policies and compliance rules.

- Explainable AI Enhancement – More detailed explanations can be provided for every risk prediction.
- OCR Integration – Scanned documents and images can be processed using Optical Character Recognition.
- Cloud Deployment – The system can be deployed as a secure cloud-based service.
- Continuous Learning – User corrections can be used to improve future classification accuracy.
- Advanced Compliance Monitoring – The system can support different industry-specific compliance frameworks.
- Integration with Enterprise Systems – It can be connected with document management, workflow, and enterprise applications.

14. Conclusion

The AI Document Risk & Compliance Classifier provides an intelligent approach for automatically analyzing and categorizing business documents based on potential risk and compliance characteristics. By combining Artificial Intelligence, Natural Language Processing, and Machine Learning, the system reduces the effort required for initial document review.

The use of confidence scoring, explainability, validation, and manual review makes the system more useful and reliable by ensuring that uncertain predictions can be examined by human reviewers. The interactive dashboard provides a clear overview of document classifications and risk patterns.

Overall, the proposed system can help organizations save time, improve document review consistency, identify potential risks earlier, and support better compliance management. With future enhancements such as OCR, multilingual NLP, advanced AI models, and enterprise integration, the system can become a more powerful document risk-management solution.

15 REFERENCE

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Vaswani, A. et al., Attention Is All You Need, Advances in Neural Information Processing Systems, 2017.

Python Documentation

Scikit-learn Documentation

spaCy Documentation

Streamlit Documentation

Hugging Face Transformers Documentation

